MathWorks

MATLAB EXPO ONLINE

Consistent Development Environments with MATLAB Projects and Package Management

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Agenda

10 min	MATLAB Projects and MATLAB Package Manager Overview
70 min	Hands-on Workshop
10 min	Q&A

Feel free to ask questions in the Q&A at any time during the workshop



R2026b provides tools to build your software in MATLAB and Simulink



MATLAB Projects

Consistent Development Environment



MATLAB Package Manager

Reliable Distribution

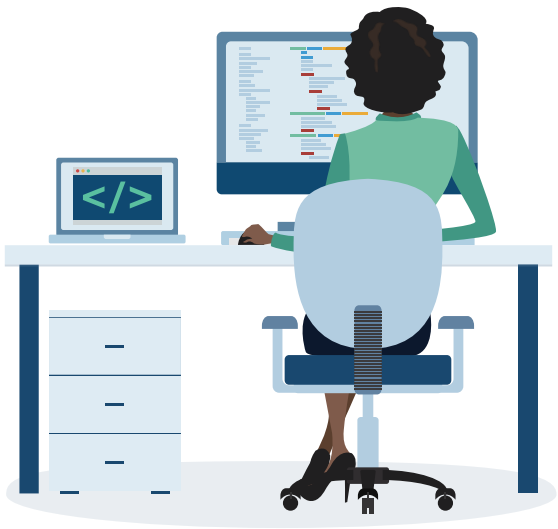
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MATLAB Projects

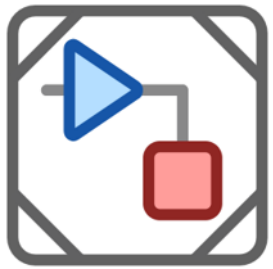
Consistent Development Environment

Your code needs to work everywhere



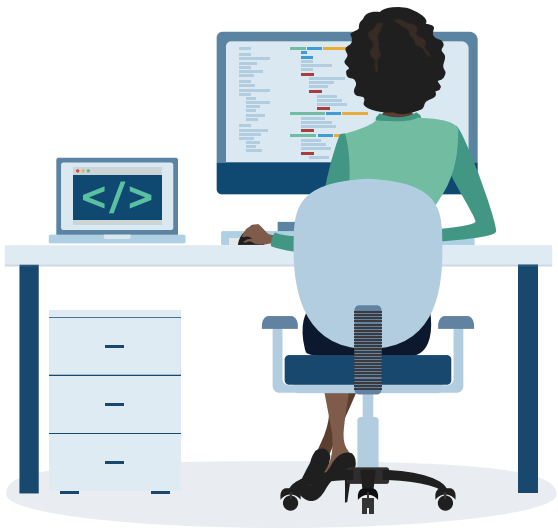


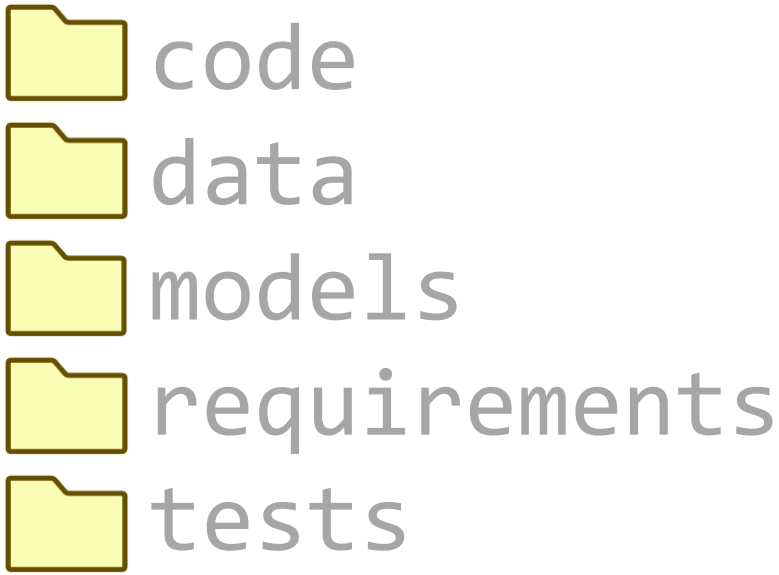






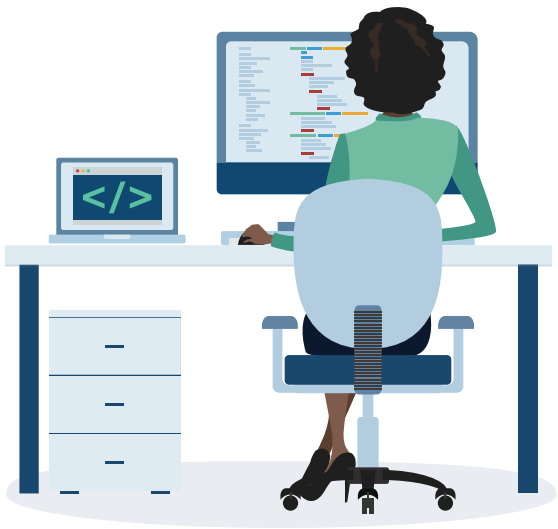
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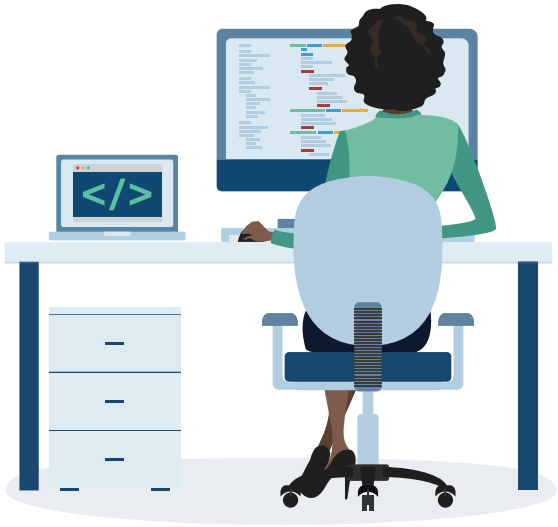
???

Your code needs to work everywhere



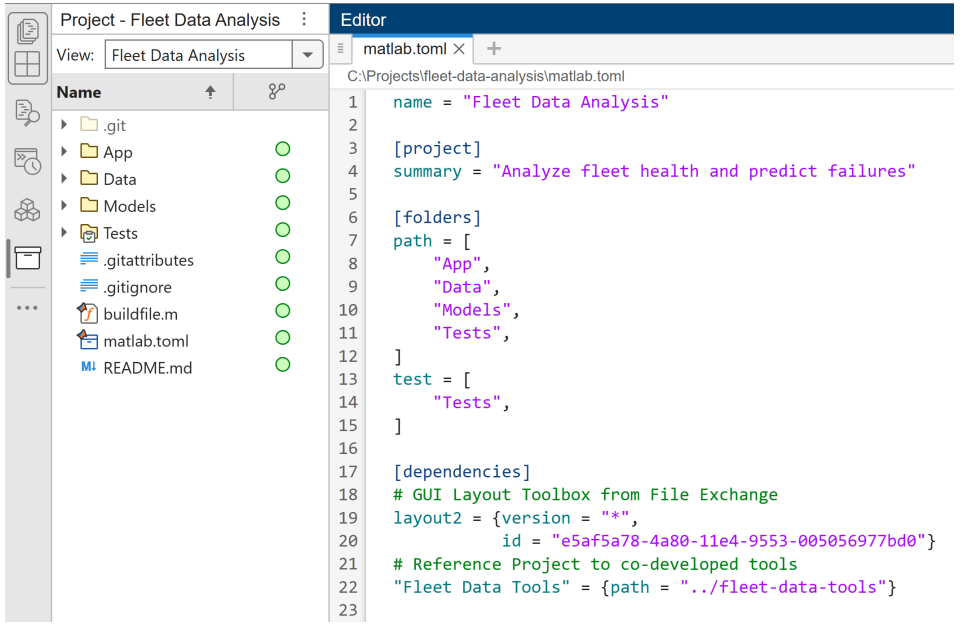
- code
- data
- models
- requirements
- tests
- matlab.toml

Your code needs to work everywhere

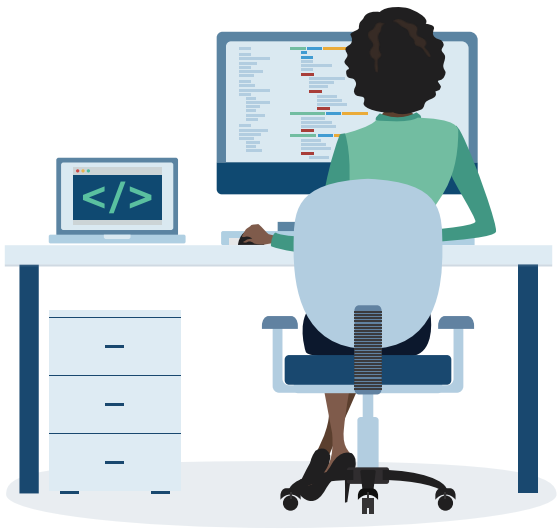




MATLAB Projects



Your code needs to work everywhere



Project - Fleet Data Analysis

View: Fleet Data Analysis

Name

- .git
- App
- Data
- Models
- Tests
- .gitattributes
- .gitignore
- buildfile.m
- matlab.toml
- README.md

Editor

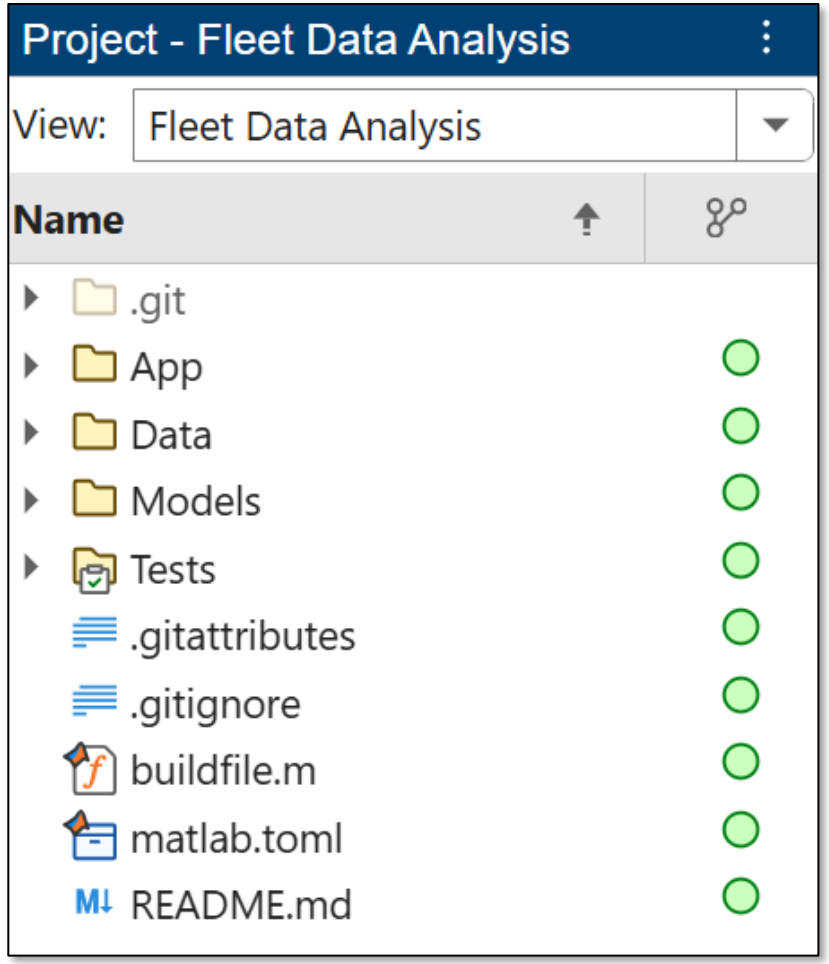
matlab.toml

C:\Projects\fleet-data-analysis\matlab.toml

```
1  name = "Fleet Data Analysis"
2
3  [project]
4  summary = "Analyze fleet health and predict failures"
5
6  [folders]
7  path = [
8      "App",
9      "Data",
10     "Models",
11     "Tests",
12 ]
13 test = [
14     "Tests",
15 ]
16
17 [dependencies]
18 # GUI Layout Toolbox from File Exchange
19 layout2 = {version = "*",
20             id = "e5af5a78-4a80-11e4-9553-005056977bd0"}
21 # Reference Project to co-developed tools
22 "Fleet Data Tools" = {path = "../fleet-data-tools"}
23
```

MATLAB Projects – Consistent Development Environment

- Consistent path setup across machines
  - Your machine, my machine, and in CI
- Defined with single **`matlab.toml`** file R2026b
- Package dependency installation R2026b
- Configure the code and models distributed in a package R2026b



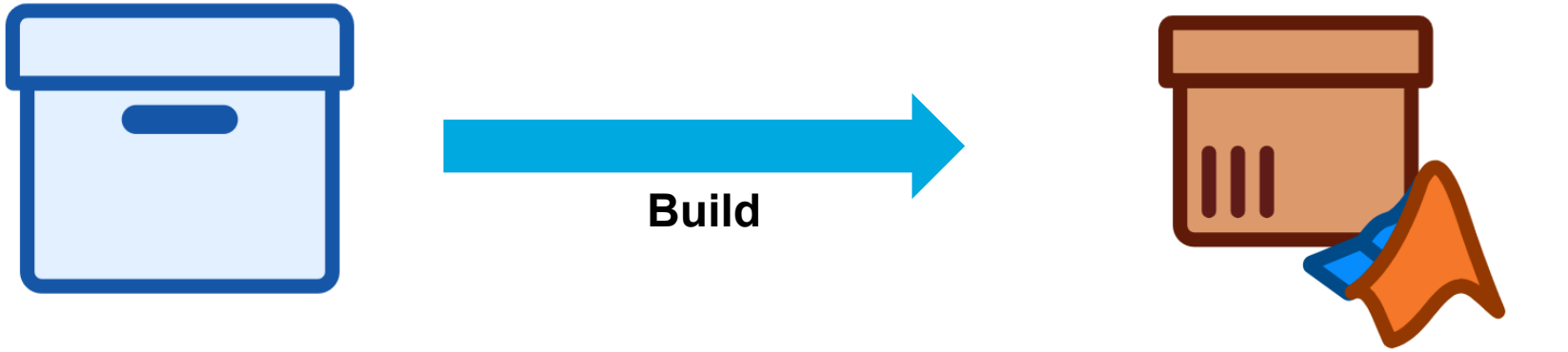
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MATLAB Projects

Consistent Development Environment

R2026b provides tools to build your software in MATLAB and Simulink



MATLAB Projects

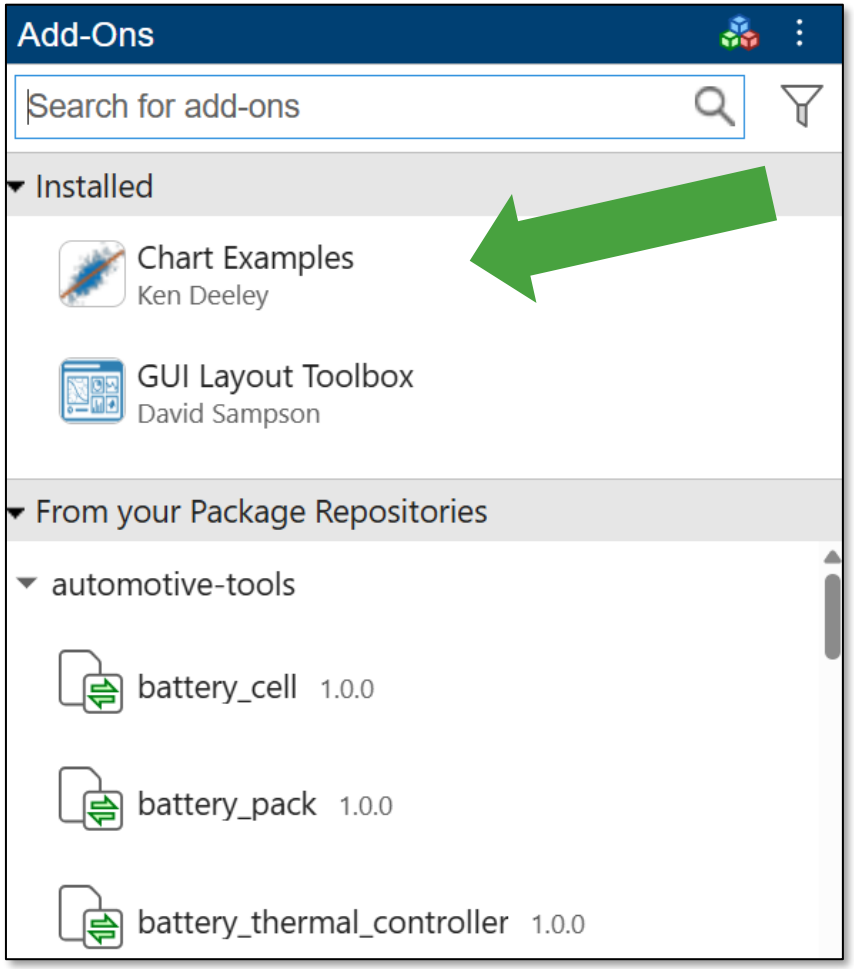
Consistent Development Environment

MATLAB Package Manager

Reliable Distribution

MATLAB Package Manager distributes your vital software in R2026b

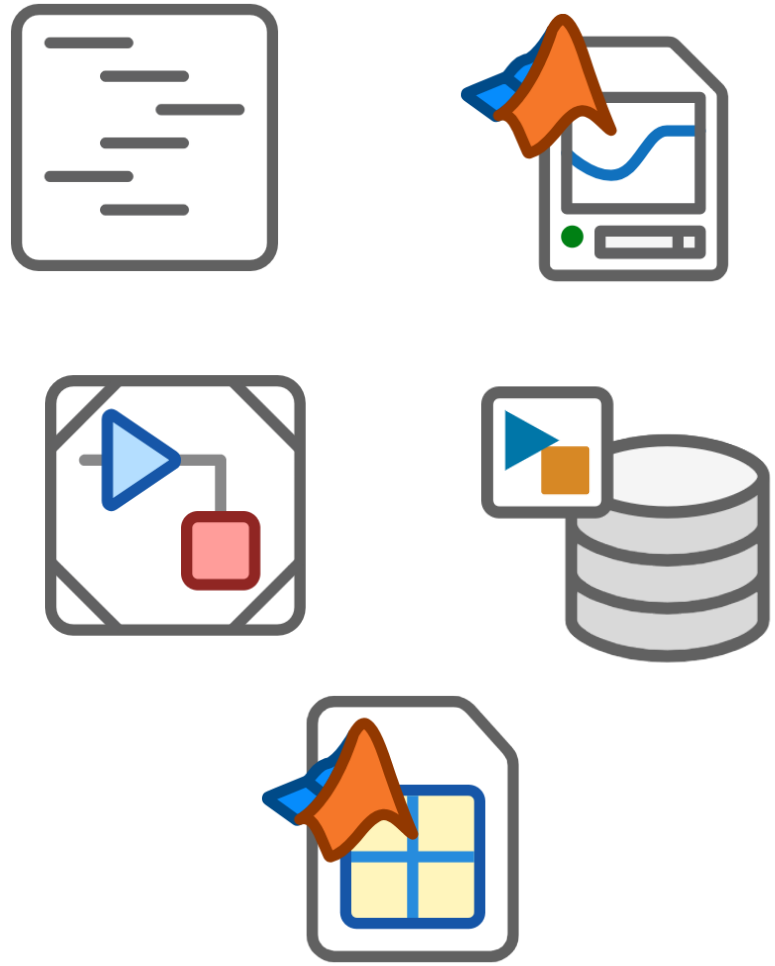
- Install from File Exchange, shared folders, and more
- **`mpm*`** APIs for scriptable installations
- Integration with MATLAB Project startup and shutdown
- Store packages as **`.mltbx`** files



Support for additional 3rd Party Package Repositories through new MathWorks Reference Architecture: https://github.com/mathworks-ref-arch/ref-impl-matlab-package-repo-service

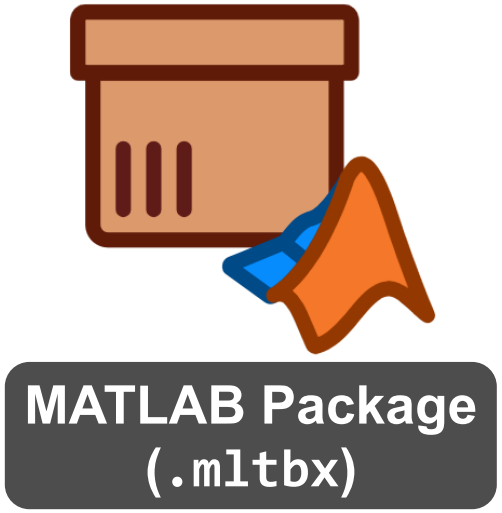
What can go into MATLAB Packages?

- MATLAB Code
- Apps
- Data
- Simulink Models
- Simulink Data Dictionaries
- … Anything you want on the MATLAB Path



What can go into MATLAB Packages?

- MATLAB Code
- Apps
- Data
- Simulink Models
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- … Anything you want on the MATLAB Path



Installing a package from File Exchange using MPM

- Explore File Exchange to find a package to install
- Copy the **`mpminstall`** command
- Install the package and any dependencies



Command Window

```
>> mpminstall Chart_Examples

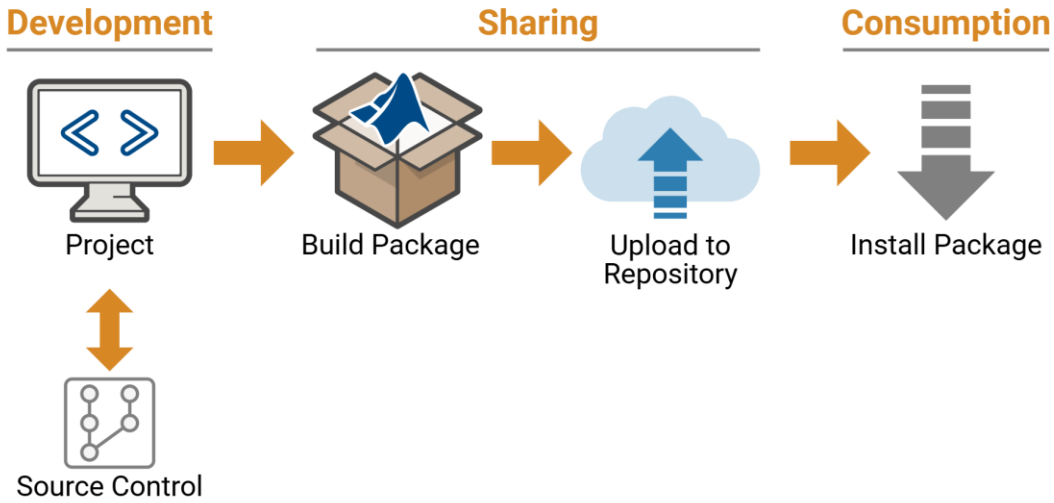
You will be accessing content made available under separate license terms.
License terms are available in the package installation folder.

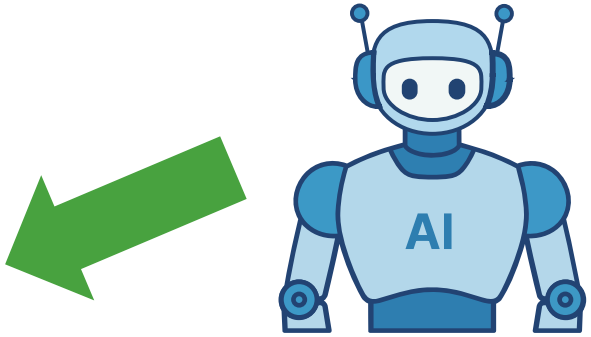
The following packages will be installed:
    Chart_Examples@2.3.2  (Repository: File Exchange Repository)
        docmaker@0.7.0  (Repository: File Exchange Repository)
        test_framework_extensions@1.0.7  (Repository: File Exchange Repository)

Do you want to continue? [YES/no]:
```

Additional Resources

- Develop and Share Software in MATLAB Overview
- MATLAB Projects TOML Format (.toml)
- MATLAB build tool new PackageTask
- Install from File Exchange
- Share and install packages with custom repositories using MATLAB package repository service
- R2026b Release Notes for Software Development
- MATLAB Software Development Skills (in MATLAB Agentic Toolkit)





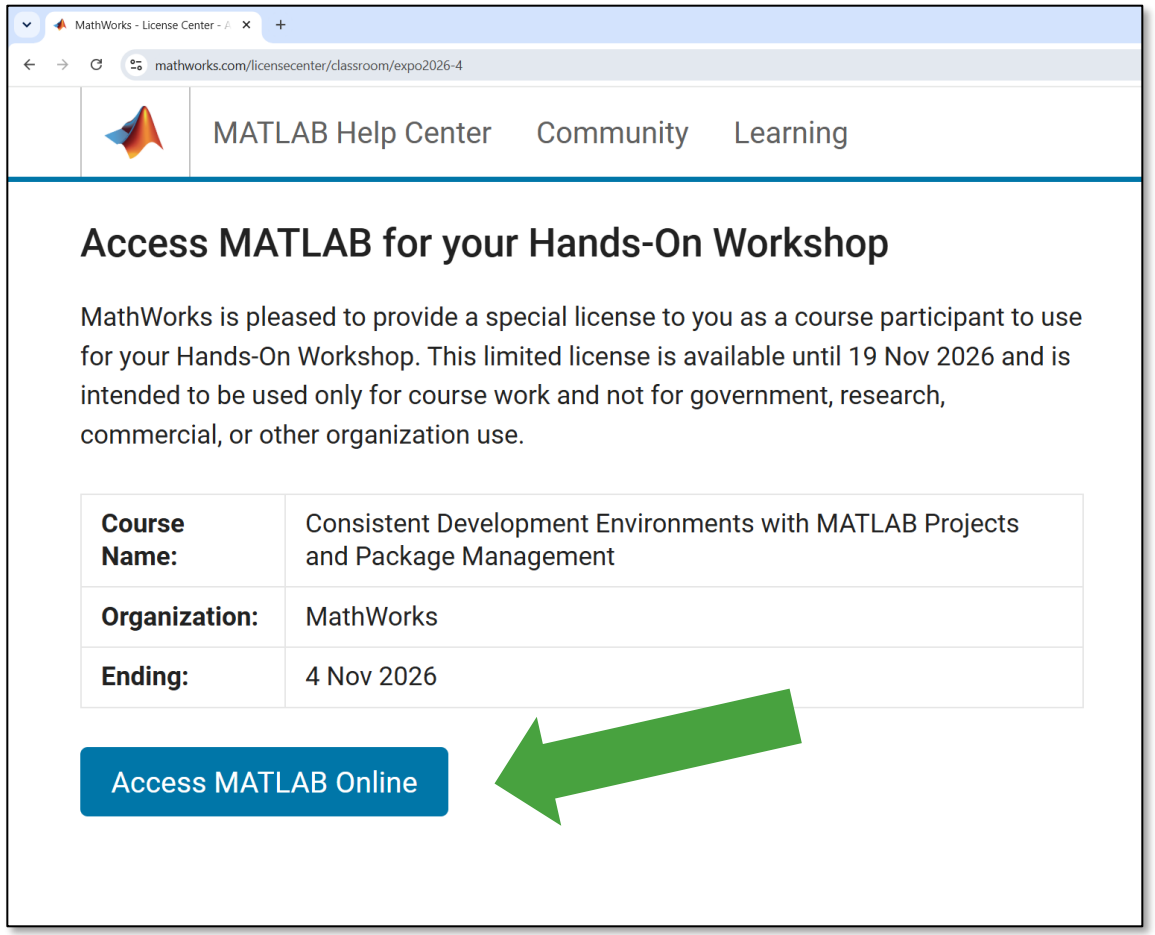
Agenda

10 min	MATLAB Projects and MATLAB Package Manager Overview
70 min	**Hands-on Workshop**
10 min	Q&A

Complimentary Workshop License

Today we will be using R2026b in MATLAB Online

- Complimentary license: https://www.mathworks.com/licensecenter/classroom/expo2026-4
- Products used today
  - MATLAB
  - Statistics and Machine Learning Toolbox



Workshop Steps

Open in MATLAB Online

1. Click Open MATLAB Online in GitHub Repository



2. Open the MATLAB Project



3. Add a package dependency



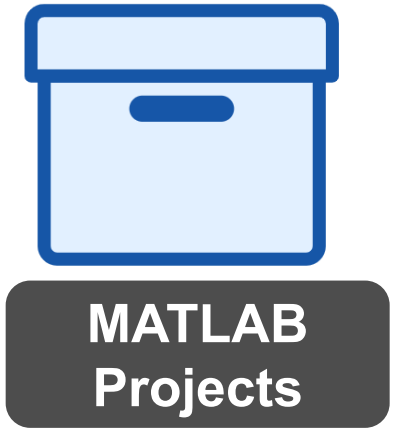
4. Build the package



5. Install the new package with dependencies

Follow along with the WorkshopGuide.m Live Script







Let's get started with the hands-on workshop!

https://github.com/mathworks/MATLAB-Expo-2026-Projects-and-Package-Management

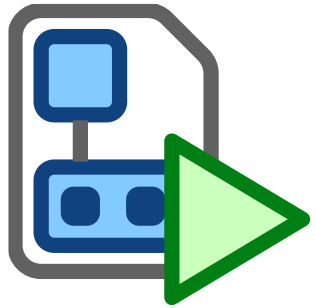
Open in MATLAB Online

Workshop Summary



MATLAB Projects

Consistent Development Environment



MATLAB Build Tool

Build Automation

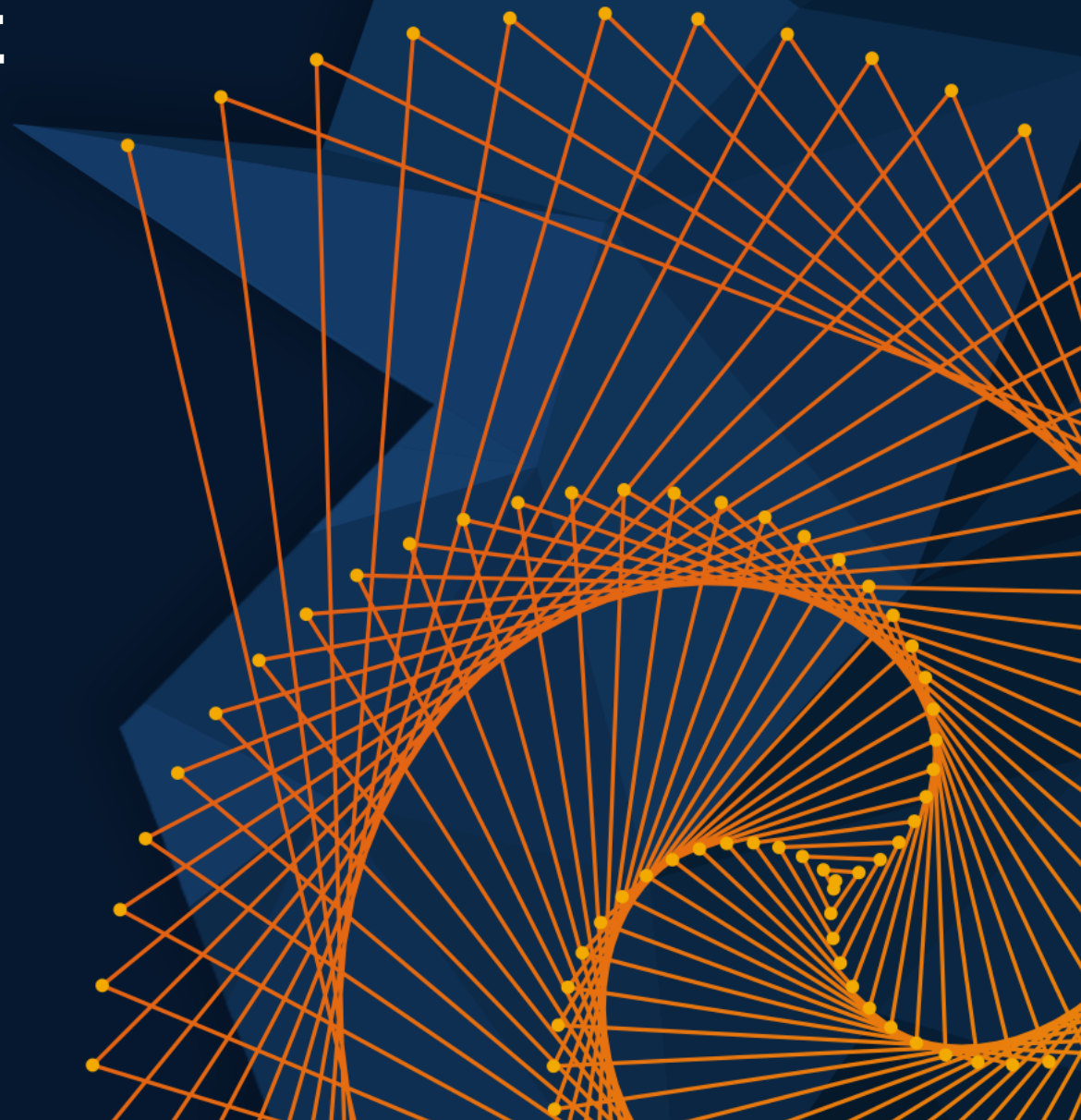


MATLAB Package Manager

Reliable Distribution

MATLAB EXPO ONLINE

Thank you!



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